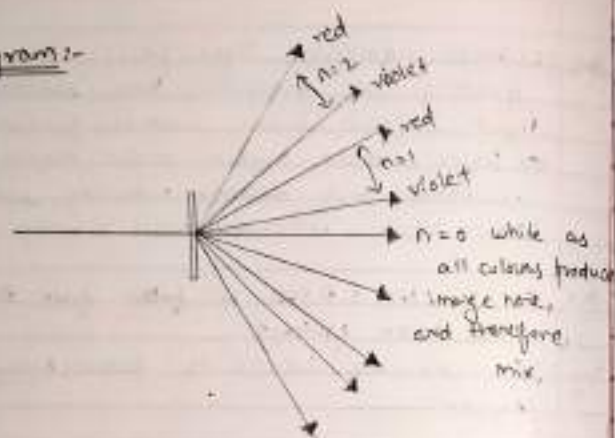


Diagram:-



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Experiment No. 8

Object:- To determine the coherence length and coherence time of laser using He-Ne laser.

Apparatus:- Spectrometer, laser source, optical screen, double convex lens, optical slit, diffraction grating & one convex lens with lens holder.

Theory & Formula:- Coherence is an important property of light it means that the co-ordinate motion of several waves maintaining a fixed and predictable phase relationship with each other. It is of two types:

- Temporal coherence
- Spatial coherence

If the phase difference ($\Delta\phi$) is measured in single point in the space at the beginning & end of a fixed time interval (Δt) & does not change with time then waves are said to possess temporal coherence.

Coherence time & Coherence length L_c :

$$L_c = c \tau_c = c / \Delta \nu = \lambda^2 / \Delta \lambda$$

Calculation:-

The wavelength λ of any line is given by

$$\lambda_1 = \frac{(0.46) \sin \theta_1}{n} \text{ cm}$$

$$n=1$$

$$\lambda_1 = (0.46) \sin \theta_1 = 1.693 \times 10^{-4} \times \sin(12.388) \text{ cm}$$

$$= 14055 \text{ \AA}$$

$$\text{Now } \lambda_2 = \frac{(0.46) \sin \theta_2}{2} = \frac{1.693 \times 10^{-4} \times \sin(20.566)}{2} \text{ cm}$$

$$n=2$$

$$= 3665 \text{ \AA}$$

Reverse

$$\lambda_{ave} = \frac{\lambda_1 + \lambda_2}{2} = \frac{14055 + 3665}{2} = 5135 \text{ \AA}$$

So

$$\Delta \lambda = 14055 + 3665 = 17720 \text{ \AA}$$

here $0.46 = 2.54 \text{ cm}$ (grating constant)

$N = \text{No. of lines per inch}$

$\theta = \text{angle of diffraction}$

$n = \text{order of spectrum}$

1.

$$\therefore L_c = \frac{\lambda^2}{\Delta \lambda}$$

$$L_c = \frac{(5135)^2}{17720} = 1523.02 \text{ \AA}$$

By this $\therefore L_c = c \cdot \tau_c$

$$\tau_c = \frac{L_c}{c} = \frac{1523.02 \times 10^{-10}}{3 \times 10^8} = 507.67 \times 10^{-18} \text{ Sec}$$

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Where,

$c = \text{velocity of light}$

$\Delta \lambda = \text{band width of the wave packet}$

Observation table

	M.S	V.S	T.R (A)
V_1	165	1	165.016
V_2	345	8	345.183

Order of Spectra	Velocity	Central Max	Spectrum on the left of the direct image			Spectrum on the right of the direct image			$\theta_1 = A - C$	$\theta_2 = A + C$
		(A)	M.S	V.S	T.R (B)	M.S	V.S	T.R (C)		
1.	V_1	A, 165.016	161.5	1	161.516	321.5	19	321.816	23.5	245.2
	V_2	345.183	182	1	181.016	3	6	345.266	263.283	335.86
									∇	∇
						(1)			113.63	263.53
0.	V_1	A, 165.016	126.75	15	126.75	243	5	243.083	30.266	231.91
	V_2	345.183	215	1	215.016	35	12	35.90	223.223	50.06
									∇	∇
						(2)			134.07	141.40
									$\theta_1 = 123.88$	
									$\theta_2 = 205.66$	

Teacher's Signature

Teacher's Signature

Observation

1. Least count of the spectrometer = $\frac{1}{60}$
2. Grating element $(a+b) = \frac{1''}{N} = 1.693 \times 10^{-4}$

Result:-

The wavelength of the He-Ne LASER is
 $= 5195 \text{ \AA}$

Coherence length is $= 1523.02 \text{ \AA}$

Coherence time is $= 507.67 \times 10^{-18} \text{ sec}$

Precautions:-

- (1) Height of all the sliders must be same & they are aligned on the same common axis.
- (2) Do not see laser directly.
- (3) Reading of both verniers scales at both windows is essential.

Industrial Application:- He-Ne laser is used widely in commercial and industrial

applications and commonly used in laboratory demonstration of optics. It is the most common laser used in holography. Other major applications of He-Ne laser include-

- Barcode scanners
- Tool alignment



- Non-contact measuring & monitoring
- Blood analysis
- Particle counting & food testing
- Alignment of high power CO₂ & YAG treatment lasers & printing beams.

Viva Questions:-

Q1 What is LASER?

Ans It is a device to produce a strong monochromatic, collimated & highly coherent beam of light.

Q2 What is meaning of LASER?

Ans The LASER word is an acronym for "Light Amplification by Stimulated Emission Radiation".

Q3. What is the principle of LASER?

Ans The LASER works on the principle of Stimulated emission or induced emission. It emits coherent light by population inversion.

Q4 What is the difference between mercury source & Laser source?

Ans Comparison between mercury source & laser source.
(a) Light emitted from mercury light source is all direction, while, laser source produce light emitting in only one direction.

- (b) When mercury light passes through the grating it produces line spectrum while light from laser sources produces spots.
- (c) Laser source higher order maxima can be observed but mercury source has limit up to second order maxima.

Q5 Is He-Ne laser four energy level laser system?
Ans Yes, He-Ne laser is four energy level laser.

Q6 Give an example of three level laser.
Ans Ruby laser.

Q7 Define the coherence time.
The average time interval for which the field remains sinusoidal (means definite phase relationship exists) is known as "coherence time" or "temporal coherence" of the light beam.

Q8 Define the coherence length.
The average length of wave trains for which the field is sinusoidal is known as "coherent length" or "longitudinal coherent length".

Q9 Mention the basic characteristics of laser light source?

Ans LASER light is monochromatic, high intense, directional & coherent.

Q10 What is wavelength of light emitted from He-Ne laser?

Ans The wavelength of light emitted by He-Ne laser is $6328 \text{ \AA}$.